

DP Computer Science: A3.2.1 Database Schemas

1. Core Concepts & Learning Objectives

Learning Intentions

By the end of this lesson, students will be able to:

- Define a **database schema**
- Describe the **purpose** and **importance** of database schemas
- Explain the differences between **conceptual, logical, and physical schemas**
- Describe the **abstract definitions** of data structure and organisation at each schema level

Key Vocabulary

Term	Definition
Database schema	The overall structure and blueprint of a database; defines how data is organised and related
Conceptual schema	The highest-level, abstract view of what data needs to be stored (entities and relationships)
Logical schema	The relational structure of the database; defines tables, columns, and keys (DBMS-independent)
Physical schema	The lowest-level view; defines how data is stored on hardware (DBMS-specific)
Entity	A real-world object or concept about which data is stored (e.g., Customer, Order)
Relationship	An association between entities (e.g., Customer places Order)
Attribute	A property of an entity (e.g., Customer has Name, Address)
Abstraction	Focusing on essential details while ignoring irrelevant details; moving from broad ideas to specific implementations

ATL Skills

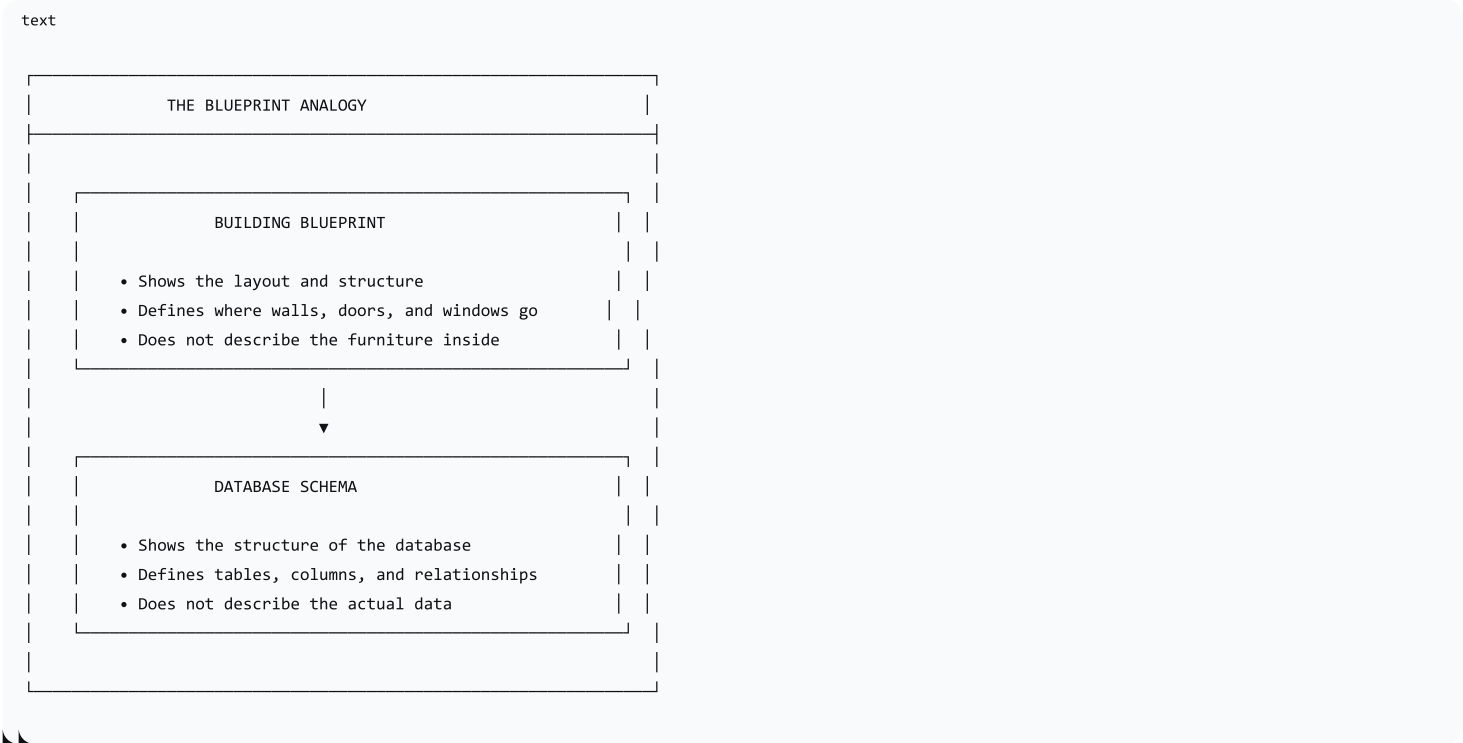
Skill	Description
Thinking	Abstraction (focusing on essential characteristics); conceptual understanding
Communication	Using technical vocabulary to describe schema levels

2. Introduction to Database Schemas

What is a Database Schema?

A database schema is the overall structure and blueprint of a database. It defines how data is organised, stored, and related.

The Blueprint Analogy

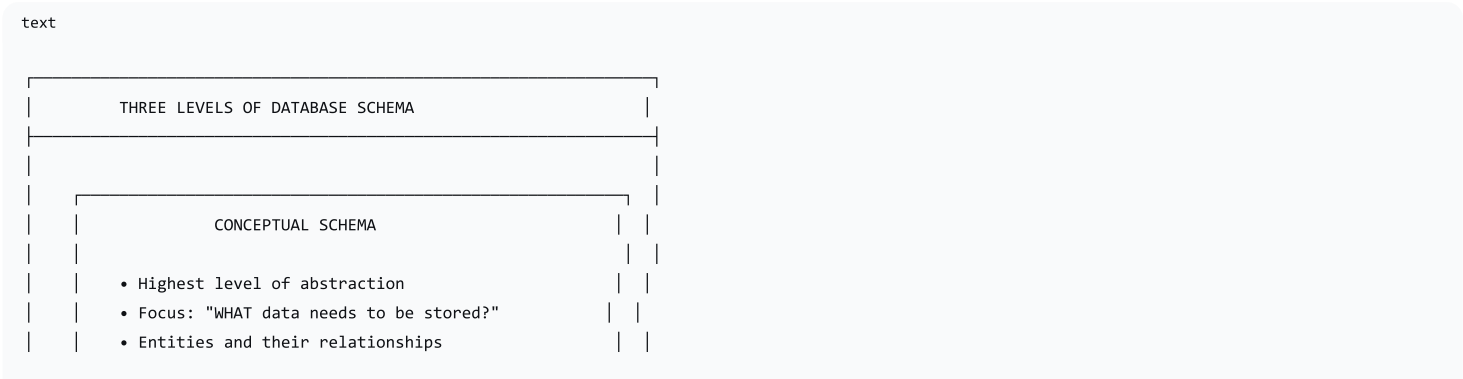


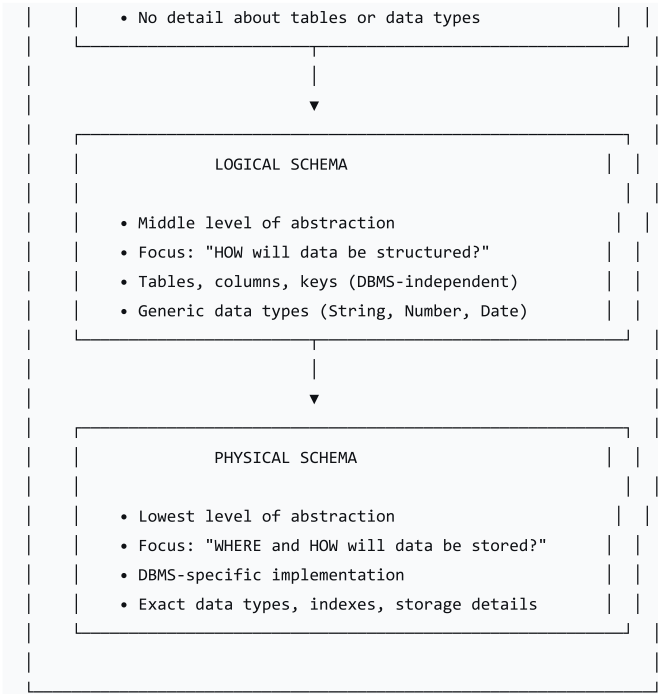
Why Are Schemas Important?

Reason	Description
Organisation	Structures data logically for efficient storage and retrieval
Integrity	Enforces rules and constraints to maintain data accuracy
Retrieval	Enables efficient querying and reporting
Data Modelling	Provides a framework for understanding data relationships
Communication	Serves as documentation for developers and stakeholders

3. The Three Levels of Abstraction

Overview





4. Conceptual Schema

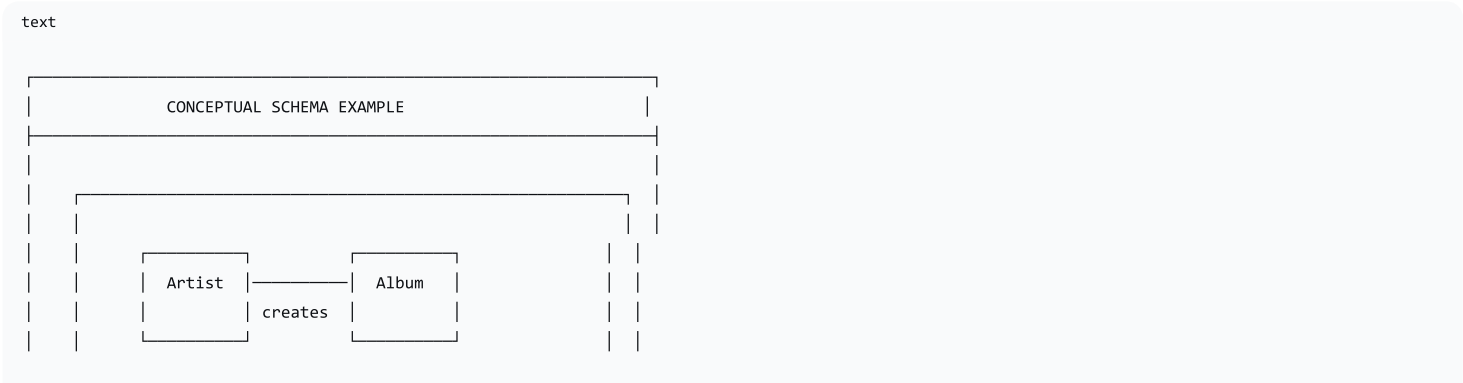
Definition

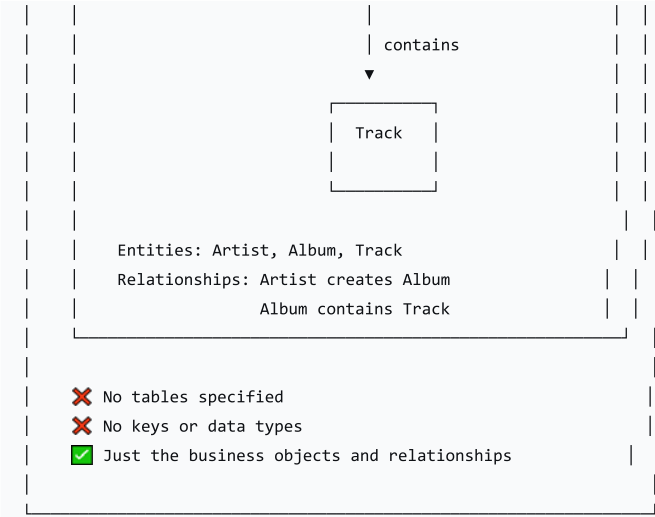
 The conceptual schema is the highest-level, abstract view of what data needs to be stored. It focuses on entities and their relationships.

Key Characteristics

Characteristic	Description
Level	Highest abstraction
Focus	WHAT data needs to be stored
Vocabulary	Entities, Relationships
Details	No tables, no keys, no data types
Audience	Business stakeholders, designers

Conceptual Schema Example: Music Streaming Library





Conceptual Schema Checklist

✓ Yes	✗ No
Entities identified	Tables defined
Relationships described	Columns specified
Business rules noted	Keys defined
	Data types mentioned

5. Logical Schema

Definition

The logical schema defines the relational structure of the database. It is DBMS-independent and focuses on tables, columns, and keys.

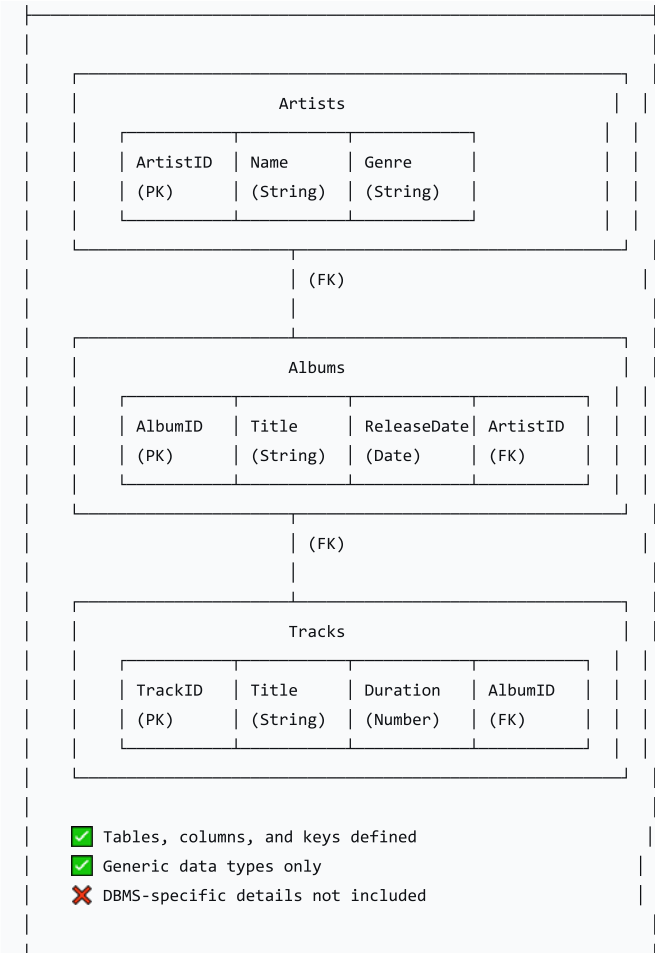
Key Characteristics

Characteristic	Description
Level	Middle abstraction
Focus	HOW data will be structured
Vocabulary	Tables, Columns, Primary/Foreign Keys
Data Types	Generic (String, Number, Date)
Audience	Database designers, developers

Logical Schema Example: Music Streaming Library

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LOGICAL SCHEMA EXAMPLE



Logical Schema Checklist

✅ Yes	❌ No
Tables defined	DBMS-specific code
Columns named	Specific storage details
Primary keys specified	Hardware considerations
Foreign keys specified	Index definitions
Generic data types used	

6. Physical Schema

Definition

 | The physical schema defines how data is stored on hardware. It is DBMS-specific and includes exact data types, constraints, and storage details.

Key Characteristics

Characteristic	Description
Level	Lowest abstraction
Focus	WHERE and HOW data will be stored
Vocabulary	Tables, indexes, partitions, constraints
Data Types	DBMS-specific (VARCHAR(100), INT, DATE)
Audience	Database administrators, engineers

Physical Schema Example: Music Streaming Library

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PHYSICAL SCHEMA EXAMPLE

DBMS: MySQL

CREATE TABLE Artists (
ArtistID INT PRIMARY KEY AUTO_INCREMENT,
Name VARCHAR(100) NOT NULL,
Genre VARCHAR(50)
);

CREATE INDEX idx_name ON Artists (Name);

CREATE TABLE Albums (
AlbumID INT PRIMARY KEY AUTO_INCREMENT,
Title VARCHAR(150) NOT NULL,
ReleaseDate DATE,
ArtistID INT,
FOREIGN KEY (ArtistID) REFERENCES Artists(ArtistID)
);

CREATE INDEX idx_artist_id ON Albums (ArtistID);

✔ DBMS-specific data types (VARCHAR, INT, DATE)

✔ Constraints (NOT NULL, FOREIGN KEY)

✔ Indexes for performance

Physical Schema Checklist

✔ Yes	✘ No
DBMS-specific data types	Generic data types only
Constraints (NOT NULL, UNIQUE)	
Index definitions	
Storage details (tablespaces, partitions)	



Yes



No

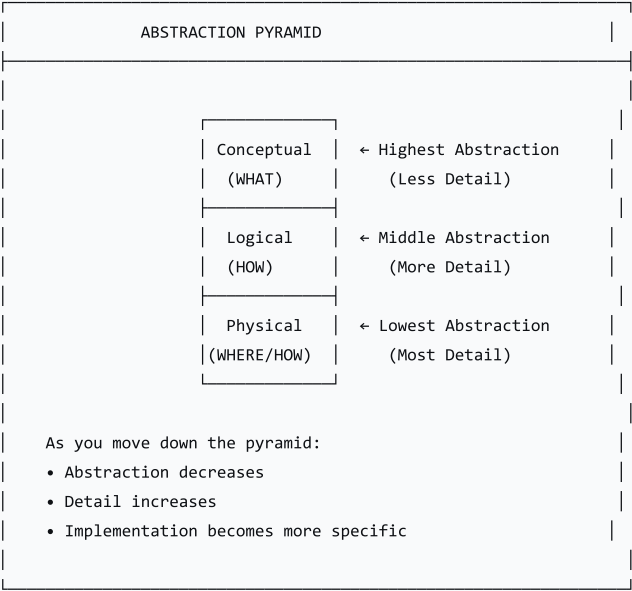
Hardware considerations

7. Summary: Comparing the Three Schemas

Feature	Conceptual	Logical	Physical
Abstraction Level	Highest	Middle	Lowest
Focus	WHAT	HOW	WHERE/HOW
Key Terms	Entities, Relationships	Tables, Keys, Generic Types	DBMS-specific code, Indexes
Data Types	None	Generic (String, Number)	Specific (VARCHAR, INT)
DBMS Dependent?	No	No	Yes
Audience	Stakeholders, Designers	Designers, Developers	DBAs, Engineers
Example	Artist → Album → Track	Tables with FK relationships	CREATE TABLE ...

The Abstraction Pyramid

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8. Activity: Music Streaming Library

Instructions

1. Design a database schema for a **Music Streaming Library**
2. Describe the database at all **three levels** of abstraction
3. Do NOT use DBMS-specific code in the logical design

Conceptual Schema

Focus: Entities and relationships

Entity	Description
Artist	A musician or band
Album	A collection of tracks
Track	A single song

Relationships:

- An **Artist** creates one or many **Albums**
- An **Album** contains one or many **Tracks**

Logical Schema

Focus: Tables, columns, keys (generic data types)

Table	Columns	Keys
Artists	ArtistID (PK), Name (String), Genre (String)	ArtistID
Albums	AlbumID (PK), Title (String), ReleaseDate (Date), ArtistID (FK)	AlbumID, ArtistID
Tracks	TrackID (PK), Title (String), Duration (Number), AlbumID (FK)	TrackID, AlbumID

Physical Schema

Focus: DBMS-specific implementation (MySQL)

```
sql

-- Artists Table
CREATE TABLE Artists (
  ArtistID INT PRIMARY KEY AUTO_INCREMENT,
  Name VARCHAR(100) NOT NULL,
  Genre VARCHAR(50)
);

-- Albums Table
CREATE TABLE Albums (
  AlbumID INT PRIMARY KEY AUTO_INCREMENT,
  Title VARCHAR(150) NOT NULL,
  ReleaseDate DATE,
  ArtistID INT,
  FOREIGN KEY (ArtistID) REFERENCES Artists(ArtistID)
);

-- Tracks Table
CREATE TABLE Tracks (
  TrackID INT PRIMARY KEY AUTO_INCREMENT,
  Title VARCHAR(150) NOT NULL,
  Duration INT, -- Duration in seconds
  AlbumID INT,
  FOREIGN KEY (AlbumID) REFERENCES Albums(AlbumID)
```



```
);

-- Indexes for performance
CREATE INDEX idx_artist_name ON Artists(Name);
CREATE INDEX idx_album_title ON Albums(Title);
```

9. Lesson Sequence

Lesson Plan (60 minutes)

Phase	Time	Activity
Introduction	10 min	"Schema Brainstorm" — think about organisation in a library, school, or shop
Three Levels	20 min	Presentation: Conceptual, Logical, Physical schemas; abstraction pyramid
Activity	20 min	Music Streaming Library — groups design the schema at all three levels
Assessment	10 min	Share findings; eBook quiz; recap key takeaways

Activity: Schema in Action

Stage	Task	Time
Stage 1	Conceptual schema: Identify entities and relationships	5 min
Stage 2	Logical schema: Define tables, columns, and keys (generic types)	5 min
Stage 3	Physical schema: DBMS-specific implementation (data types, constraints, indexes)	5 min
Peer Review	Groups pair up to review Logical vs. Physical choices	5 min

10. Assessment Activities

Assessment Type	Description
Class Participation	Observe engagement and contributions
Activity	Evaluate ability to design schemas at all three levels
Quiz	eBook quiz or Quizlet

Sample Quiz Questions:

1. What is a database schema?
2. What is the difference between a conceptual and a logical schema?
3. What data types are used in a logical schema?
4. Which schema level includes DBMS-specific code?

11. Differentiation

Level	Strategy
Support	Provide visual aids and scaffolds; use guided examples
Challenge	Add complex relationships; explore advanced indexing strategies

12. Resources

- **Presentation** (Slide Deck)
- **eBook** (A3.2.1 with quiz)
- **Quizlet** (A3.2.1 vocabulary flashcards)

13. Summary: Key Takeaways

Schema Level	Focus	Key Terms	DBMS Dependent?
Conceptual	WHAT data is stored	Entities, Relationships	No
Logical	HOW data is structured	Tables, Keys, Generic Types	No
Physical	WHERE/HOW data is stored	Specific Types, Indexes, Constraints	Yes

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KEY REMINDERS

✔

Conceptual = The idea (entities and relationships)

✔

Logical = The independent blueprint (tables, keys)

✔

Physical = The implementation (DBMS-specific code)

✔

Abstraction decreases as you move down the pyramid

✔

Detail increases as you move down the pyramid

“The schema is the blueprint of the database—it defines the structure and relationships before any data is stored.”